

Marginal-price support and repeated optimization of electric-bus fleets

Research draft 0.6 for author review — 28 September 2026

Abstract

Electric-bus charging combines continuous energy decisions with indivisible service assignments. We study whether a complete feasible fleet schedule is a best response to the marginal electricity prices generated by its own load. An exact, fully replenished two-service construction exhibits a positive gap between physical and convexified planning, with a robustness neighborhood allowing charging losses, a reserve and a finite connector. In a replication family, the absolute planning gap vanishes while whole-operator regret remains positive along a subsequence; regret per reserved pair vanishes. Computational development studies then examine repeated complete-fleet optimization on two synthetic cases and two depot variants of one public timetable. Reusing feasible fleets and physical-pricing bounds improves particular bounds, but public optimality gaps remain open and native starting-fleet effects are mixed. We separate feasible dispatch, convex mixtures, numerical certification and paid computation. These results motivate further proposal-based acceleration without establishing a speedup or a learned scheduling method.

Keywords: electric-bus scheduling; convex-hull pricing; marginal tariffs; charging coordination; repeated optimization.

1 Introduction

An electric-bus operator must cover mandatory passenger services while assigning vehicles and scheduling charging. A tariff can shift charging toward cheaper periods, but a fleet large enough to affect aggregate electricity demand also changes the marginal cost behind that tariff. Even with continuously adjustable charging, its service assignments remain indivisible. The resulting question is whether a physically realizable fleet schedule minimizes the operator’s bill when evaluated at the marginal prices associated with that schedule’s load.

We study this question using complete fleet plans. Each plan includes all services, vehicle movements, charging and operating cost. Convexifying their joint cost and load vectors provides a useful planning lower bound, but a point in that hull is generally not an executable daily dispatch. Charging a convex supply cost at the mean load also differs from taking the mean cost of randomized physical dispatches. Maintaining these distinctions is essential both for interpreting a pricing failure and for assessing computational shortcuts.

The contribution has two parts. First, a small continuous-charging construction makes the obstruction explicit under full terminal replenishment. A perturbation permits a positive reserve, charging losses and one finite connector. A replication calculation then separates an aggregate planning gap from the incentives of a whole-fleet operator and of participants with individually

reserved resources. These are application-specific exact results within established nonconvex pricing theory, rather than a new general decentralization theorem. Second, development computations ask what repeated optimization can reuse, and what still requires global physical-fleet pricing. Their tables expose both useful bounds and incomplete solves.

The computational scope is deliberately stated at the level observed: two small synthetic diagnostics and two depot variants of one public timetable. The campaigns share cases and some source evidence; they are not independent replications across transport networks. Their clearest findings concern particular bound improvements and the cost of fresh physical pricing. They do not demonstrate public-case global optimality, a general speedup at equal quality, or learned route prediction. This manuscript therefore provides an exact mechanism and a computational foundation for a broader study, with the remaining empirical questions explicit.

2 Related work

Electric-bus scheduling combines discrete service and vehicle assignments with battery and charging decisions. Wu et al. study a multi-depot bus-scheduling problem with time-of-use tariffs and peak-load objectives, solved with branch-and-price and trip-chain reuse [Wu et al., 2022]. De Vos et al. include partial charging and limited station capacity in a path-based public-transit formulation [de Vos et al., 2024]; Parmentier et al. develop scalable route and recharge scheduling by column generation for electric fleets serving passenger requests [Parmentier et al., 2023]. These works establish the operational role of duty construction, energy limits, and charging resources. The question here is different: how a complete feasible fleet plan relates to the convex hull of complete plans under a common convex supply-cost model. A relaxation whose columns are individual vehicle duties need not equal this whole-fleet hull, and a mean load in the hull need not be executable as one day’s bus schedule.

Public timetable and movement data make this distinction testable, while leaving modeling choices visible. Sistig et al. generate vehicle and crew schedules for twenty German transport networks from public GTFS inputs; their accompanying versioned dataset includes routes, trips, stops, possible deadheads, and schedules [Sistig et al., 2025a,b]. The present public intake uses one Hildenbrand timetable in two restricted depot variants. Source distances and times are combined with declared energy, battery, charger, and synthetic supply-cost assumptions; the source supplies reference bus parameters, while EGG’s reserve, charger, horizon and convex supply-cost assumptions remain declared scenario choices. It supplies no EGG market-supply function. The reviewed computational campaigns remain small development studies on this one timetable and its two depots. Their public global bounds remain open, and they establish neither an equal-quality speedup nor a learned-proposal benefit.

Nonconvex electricity-market research explains why marginal energy prices can fail to support a dispatch selected with indivisible operating decisions. Gribik et al. compare pricing rules for unit commitment and dispatch and discuss the energy uplift needed when no energy-price vector supports the selected quantities [Gribik et al., 2007]. O’Neill et al. develop prices from a mixed-integer model and an associated linear program for nonconvex markets [O’Neill et al., 2005]. Convex-hull pricing then studies uniform prices through convexification: Schiro et al. explain its formulation and implementation issues, while Hua and Baldick give a convex primal formulation for unit-commitment hull pricing [Schiro et al., 2016, Hua and Baldick, 2017]. We use this established convexification and supporting-price logic for a stated fleet set and supply-cost function. The resulting own-price best-response test is conditional on that common model; it is neither a general

market-clearing theorem nor a settlement design. In particular, fleet regret at its own marginal price is not an uplift payment under any of these market rules.

Aggregation results also require their own scale and participant definitions. Starr’s classical quasi-equilibrium result gives an aggregation perspective for economies with nonconvex preferences [Starr, 1969]. Our replication construction is proved directly for its specified fleet schedules: the aggregate planning gap tends to zero while whole-operator regret can remain positive, whereas per-bus and individually reserved-pair incentives vanish. A Shapley–Folkman aggregation argument does not by itself prove this exact replication result or force the absolute regret of one composite operator to vanish. The distinction is between aggregating many separately bounded participants and enlarging a single participant whose opportunity set remains indivisible.

Price coordination for flexible EV charging is likewise established. Gan et al. formulate each vehicle’s charging profile as a continuous, rate-bounded vector with a fixed energy requirement over its availability window, and give a decentralized price-update protocol that converges for a convex aggregate-load objective [Gan et al., 2013]. That framework coordinates charging once availability and energy needs are specified. Electric-bus schedules also choose which trips connect to which vehicles; those discrete choices alter each bus’s charging windows and the joint feasible load set. Convex coordination over charging profiles therefore does not imply that a discrete complete bus schedule is supported by its own marginal price.

Finally, fixed-schedule fleet counting has a classical analogue in Dantzig and Fulkerson’s minimum-tanker problem for meeting a prescribed schedule [Dantzig and Fulkerson, 1954]. Fixed-schedule fleet-count reductions address how many vehicles cover fixed movements. By themselves, they do not certify battery-state feasibility, shared-charger feasibility, or price support. We keep the minimum-fleet flow diagnostic separate from complete-schedule charging replay and from the economic comparison between physical plans and their convex hull.

3 Complete fleets and marginal-price support

The unit of optimization is a *complete* fleet plan, rather than an individual vehicle duty. Fix a timetable, allowed directed movements, a charging horizon, and the equipment and energy assumptions. Let $\mathcal{F} \neq \emptyset$ contain precisely the plans that cover every mandatory service once, assign the services to used buses, route those buses through allowed movements, respect battery and charger constraints, and restore every used bus to its required terminal inventory. For $x \in \mathcal{F}$, write $c(x)$ for intrinsic operating cost and $e(x) \in \mathbb{R}_+^T$ for grid energy bought in the T market intervals. An unused bus contributes neither charging nor cost. We use the same $\mathcal{F}$, c , and e for physical planning, price response, and convexification. Finitely many discrete duties with bounded continuous charging give a compact $\mathcal{F}$ in the cases below; assume more generally that its image $\{(e(x), c(x)) : x \in \mathcal{F}\}$ is compact.

In the timed physical model, each used path starts at battery inventory B . A service or directed movement consumes its declared energy in chronological order. Inventory remains between reserve r and capacity B at every event. A depot charging session buying q grid kWh adds ηq inventory, where $0 < \eta \leq 1$, within its availability window and bus-power limit. Concurrent sessions obey the shared power or connector limit. Each path finishes at inventory B by the declared horizon. In the homogeneous single-connector formulation, charging intervals are split at all visit and market boundaries. If interval k has duration d_k and power P_k , its charges obey $0 \leq z_{mk} \leq P_k d_k y_m$ and $\sum_m z_{mk} \leq P_k d_k$, where y_m indicates the selected visit. Serial allocation of z_{mk}/P_k hours to the visits realizes these totals when switching time is zero, efficiency is constant, and charging does not

taper. The worked constructions below assume zero travel cost and no unlisted travel or auxiliary load; the public case declares its movement energy separately.

Let $F : \mathbb{R}_+^T \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper, closed, convex supply cost that admits a finite differentiable convex extension to a neighborhood of the convex hull of feasible loads. The physical and complete-fleet-hull values are

$$\begin{aligned} D &= \min_{x \in \mathcal{F}} \{c(x) + F(e(x))\}, \\ CH &= \min_{(e,c) \in \text{conv}\{(e(x), c(x)) : x \in \mathcal{F}\}} \{c + F(e)\}, \quad \Delta = D - CH \geq 0. \end{aligned} \quad (1)$$

The hull mixes whole feasible plans, including their charger constraints; its mean load is a pricing and bounding object, not necessarily a dispatch. For a posted price p , define the complete-fleet response $V(p) = \min_{x \in \mathcal{F}} \{c(x) + p^\top e(x)\}$. At a physical plan x , $p_x = \nabla F(e(x))$ and

$$r(x) = c(x) + p_x^\top e(x) - V(p_x). \quad (2)$$

The alternative in $V(p_x)$ takes the posted price as fixed. This is the appropriate support test for a price-taking fleet, even though its original load determined p_x . A strategic fleet instead compares bills such as $c(y) + \nabla F(e(y))^\top e(y)$ across its alternatives y and anticipates the resulting price change. That objective differs from both $c(y) + p_x^\top e(y)$ and the planner's $c(y) + F(e(y))$; it needs a specified settlement rule and strategic game. The present price-support and compensation statements make no claim about strategic behavior [O'Neill et al., 2005, Gribik et al., 2007, Gan et al., 2013].

Proposition 1 (Complete-fleet support). *Under the common-model and attainment assumptions above, every physical $x \in \mathcal{F}$ satisfies*

$$r(x) \geq c(x) + F(e(x)) - CH \geq D - CH = \Delta. \quad (3)$$

Moreover, $D = CH$ if and only if some physical plan is a best response at its own marginal price. If $D = CH$, every physical planner optimizer has this property.

The proof in Appendix A is a specialization of classical nonconvex price-support and convexification arguments [Starr, 1969, O'Neill et al., 2005, Gribik et al., 2007, Hua and Baldick, 2017, Schiro et al., 2016]. It says nothing about whether an iterative price adjustment converges or which optimum an algorithm selects. Attainment matters in the converse: equality of infima alone does not produce a physical supporting plan.

The accounting identity also requires a stated deviation right. In this paper one fleet controls all buses and its alternative must be a member of the same $\mathcal{F}$. Separate operators using shared chargers would need reserved capacities, trades, or congestion rights before a unilateral opportunity set could be defined. For the single fleet, put $F^*(p) = \sup_{u \geq 0} \{p^\top u - F(u)\}$ and define

$$\text{LOC}_{\text{fleet}}(x; p) = c(x) + p^\top e(x) - V(p), \quad (4)$$

$$\text{LOC}_{\text{supply}}(x; p) = F(e(x)) - p^\top e(x) + F^*(p). \quad (5)$$

Whenever $F^*(p)$ is finite, both terms are nonnegative and their sum is $c(x) + F(e(x)) - [V(p) - F^*(p)]$. At a hull-optimal price and a physical planner optimizer, strong Fenchel duality makes that sum Δ ; the fleet and supply portions depend on the chosen common price. At p_x , supply LOC is zero and fleet LOC equals $r(x)$. No budget-balance or payment institution follows from this arithmetic.

For quadratic supply $F(e) = a^\top e + \frac{1}{2}e^\top Qe$ with $Q \succeq 0$, write $\|z\|_Q = (z^\top Qz)^{1/2}$. There is a useful complement to the lower bound. Let $R_Q = \max_{x,y \in \mathcal{F}} \|e(x) - e(y)\|_Q$ and $H(x) = c(x) + F(e(x)) - CH$. Then

$$H(x) \leq r(x) \leq H(x) + R_Q \sqrt{2H(x)}. \quad (6)$$

Thus a small cost gap controls own-price regret only together with the curvature-weighted range of feasible fleet loads. The proof permits singular Q and boundary loads and appears in Appendix A.

For numerical certification, valid bounds $D \in [L_D, U_D]$ and $CH \in [L_{CH}, U_{CH}]$ imply $\Delta \in [L_D - U_{CH}, U_D - L_{CH}]$. A replay-feasible physical plan supplies an upper bound on D ; a mixture of replay-feasible complete plans supplies an upper bound on CH . At any price with finite $F^*(p)$, a global price-response lower bound $\underline{V}(p) \leq V(p)$ gives the hull lower bound $\underline{V}(p) - F^*(p)$. A nearly solved restricted column pool alone cannot supply that global lower bound. In the separable implementation, $F(e) = \sum_t (a_t e_t + b_t e_t^2/2)$ and $b_t \geq 0$; the conjugate term is $[p_t - a_t]_+^2/(2b_t)$ when $b_t > 0$, and is finite at $b_t = 0$ only for $p_t \leq a_t$. These are standard convexification and Fenchel certificates over complete plans, rather than a new general convex-hull pricing method [Hua and Baldick, 2017]. Section 6 writes G_s for this supply function in market state s .

4 Exact replenished-fleet constructions

Consider two mandatory depot-to-depot services. Service A occupies hour $[0, 1]$ and service B occupies $[2, 3]$; each consumes 15 kWh. At most two identical buses may be used. Each used bus starts at its full 20 kWh capacity and must return to 20 kWh by hour 4. The early charging window is $[1, 2]$, with 10 kW per bus and in total. The terminal window is $[3, 4]$, with 30 kW per bus and in total and enough connectors. Charging is continuous and lossless; reserve, travel cost, deadhead energy, auxiliary load, switching time, and taper are zero. No charging is available outside these windows. The terminal charge is physical replenishment, so its energy and supply cost are included.

There are only two unlabeled service partitions. If one bus covers A and B, it reaches 5 kWh after A and needs at least 10 kWh before B. The early power limit fixes its early grid energy at $x = 10$ kWh; it buys 20 kWh terminally. If two buses cover one service each, the A bus buys any $x \in [0, 10]$ kWh early and $15 - x$ kWh terminally, while the B bus buys 15 kWh terminally. Every used bus returns full. Both structures therefore buy exactly 30 kWh, and initial inventories cannot subsidize the horizon.

Let a used bus cost $f = 7$ synthetic units and let grid supply cost be

$$F(x, z) = 4x + \frac{x^2 + z^2}{10}, \quad x, z \geq 0. \quad (7)$$

The one-bus physical cost is $7 + F(10, 20) = 97$. The two-bus branch has cost $14 + F(x, 30 - x)$, minimized at $x = 5$ with value 99. Yet the complete-fleet hull has a lower objective.

Proposition 2 (Replenishment does not ensure price support). *For this physical model,*

$$D = 97, \quad CH = \frac{7591}{80} = 94.8875, \quad \Delta = \frac{169}{80} = 2.1125. \quad (8)$$

The physical planner optimum has own-price regret 13. At a common hull-optimal price its fleet lost opportunity is zero and its supplier lost opportunity is $169/80$.

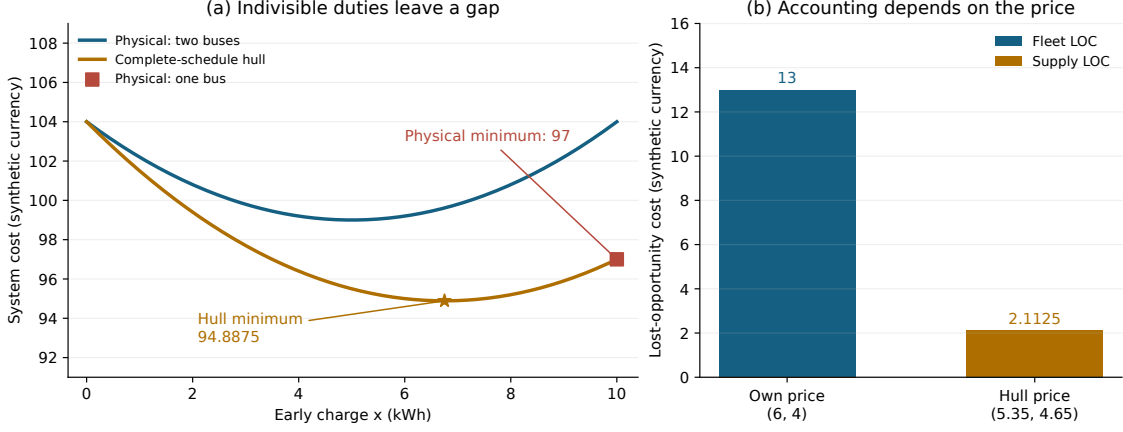


Figure 1: Exact physical branches, complete-fleet hull, and lost-opportunity accounts in the synthetic two-service case. Prices are synthetic units/kWh; the hull mean is not an executable daily fleet.

The hull optimizer mixes the one-bus plan $(x, z) = (10, 20)$ with weight $27/40$ and the two-bus plan $(0, 30)$ with weight $13/40$. Its mean early load is $27/4$ kWh. Both components are physical; their mean load and mean operating cost are a convexification, not a fractional bus schedule. Running the two plans on different days gives mean system cost 99.275 , because averaging physical supply costs differs from evaluating the convex supply cost at their mean load. The exact hull and price calculations are in Appendix A.

At the physical optimum, the own marginal price is $(6, 4)$ synthetic units/kWh. The one-bus private objective is 147 , while the two-bus plan charging entirely terminally has private objective 134 . The resulting 13 -unit regret is a fixed-price incentive to change the whole fleet. The hull price is $(5.35, 4.65)$: the two plans supporting its mean tie at private objective 153.5 . At that price, the physical one-bus optimum has no fleet lost opportunity, but the supplier term is $169/80$. These two diagnostics use different prices and answer different questions.

The binding 10 kW early limit is a fragile nominal parameter: at the same 20 kWh capacity, a positive reserve or charging loss can remove the one-bus branch. A nearby construction allows those constraints while retaining a gap. Keep two 15 kWh services, battery capacity 20 , and (7) , but use reserve $r = 1$ kWh, grid-to-battery efficiency $\eta = 19/20$, 12 kW individual and shared early power, and a single 30 kW terminal connector. The one-bus plan buys $220/19$ grid kWh early and 20 terminally, with battery states $20 \rightarrow 5 \rightarrow 16 \rightarrow 1 \rightarrow 20$. Its early power has $8/19$ kWh of headroom. The supporting two-bus plan buys $30/19$ early and 30 terminally; the two terminal sessions take $9/19$ and $10/19$ hour and fit sequentially on the single connector. Every used bus again finishes full.

Proposition 3 (A positive-gap neighborhood). *Under the modified parameters,*

$$D = \frac{38527}{361}, \quad CH = \frac{2987911}{28880}, \quad \Delta = \frac{94249}{28880} > 0. \quad (9)$$

With the 30 kW terminal connector fixed, strict feasible-headroom and optimality inequalities preserve a positive gap in an open neighborhood of the stated reserve, efficiency, and early power.

The interval and strict-inequality proof is in Appendix A. The neighborhood is conditional on this hardware model's zero switching time, constant efficiency, and lack of taper. For perspective on connector adequacy, if the original lossless case has one terminal connector of power $K \leq 30$ kW, it

is infeasible for $K < 20$, has zero gap for $20 \leq K \leq 23.5$, and has a positive gap for $23.5 < K \leq 30$. Thus the presence of one connector alone says nothing about price support.

4.1 Replication and the participant scale

Replicate A and B to n simultaneous copies each. Give the fleet n early 10 kW connectors and n terminal 30 kW connectors. Battery and service energy per bus stay fixed; supply cost scales with demand as

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}. \quad (10)$$

If m buses each cover an A–B pair, $n - m$ A-only and $n - m$ B-only buses complete the services. There are $2n - m$ used buses, and physical early energy ranges over $[10m, 10n]$. Terminal sessions have an explicit connector-feasible assignment: paired buses take m connectors, while each A-only/B-only pair can share one of the other $n - m$ terminal connectors sequentially. All buses return full.

Proposition 4 (A vanishing gap with aggregate regret). *The complete projected hull has lower operating-cost boundary $14n - 0.7E$ and optimum $CH_n = 7591n/80$. Every physical optimum takes an integer m nearest to $27n/40$; both neighboring integers are retained at a tie. With $\delta = m - 27n/40$,*

$$\Delta_n = \frac{20\delta^2}{n} \leq \frac{5}{n}. \quad (11)$$

Along $n = 40k + 1$, the whole-operator own-price regret is

$$r_n = \frac{351}{40} + \frac{169}{40n} \longrightarrow \frac{351}{40}, \quad \Delta_n = \frac{169}{80n} \longrightarrow 0. \quad (12)$$

For multiples of 40, both quantities are exactly zero.

The operator in (12) can rearrange the whole growing fleet while taking its posted price as fixed. Its regret per used bus, and regret divided by physical cost, vanish. A different institution gives each of n operators one A/B pair plus a reserved early and terminal connector. Each then chooses either the one-bus plan or a two-bus plan with $x \in [0, 10]$; these separate deviations are physically compatible under the reserved rights. At the original fleet optimum's marginal price, each pair's regret is at most $20/n$, and their sum equals the whole-fleet regret in this symmetric construction. Along $n = 40k + 1$ a dissatisfied one-bus operator has regret $13/n$, although the aggregate tends to $351/40$. The scaling example consequently distinguishes absolute cost gap, whole-operator incentive, and per-participant incentive without asserting a result for arbitrary shared-charger games. At $n = 20$, the two tied planner optima have distinct own-price regrets, 7 and 14.

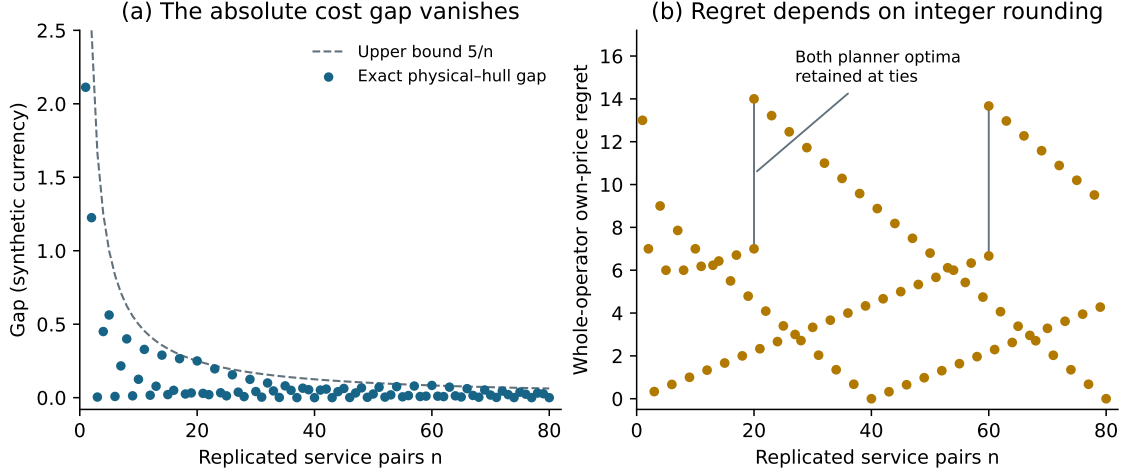


Figure 2: Exact replication for $n = 1, \dots, 80$, including both physical optima at ties. The planning gap follows the $5/n$ bound while whole-operator own-price regret varies with integer rounding. Values are synthetic units, not observations from a bus system.

5 A public timetable: feasible fleets and bounded costs

The exact two-service example isolates the mechanism; it does not measure its prevalence or size in service. For a timetable-scale check we use the published Hildenbrand instance from the Sistig electric-bus scheduling data [Sistig et al., 2025a,b]: 37 mandatory services with a complete directed deadhead matrix. We form two *separate* one-depot cases, using depot 15 or depot 16. These are declared modifications of the source network, not a reconstruction of its two-depot, multiple-output charging design. Timetable and directed travel times retain their source offsets, including the next-day services. Matrix travel times depend on origin and destination, not on departure time.

Energy and monetary inputs are explicit scenario assumptions, not observed vehicle telemetry or source objective coefficients. Service energy is 1.20 kWh/km plus 16 kWh per service hour. Directed deadheads use 0.96 kWh/km plus 16 kWh per travel hour, and explicit off-depot waiting uses 16 kWh/h; depot-parked auxiliary draw is zero. Each used bus has 400 kWh usable inventory, zero additional reserve, unit charging efficiency, and a 360 kW acceptance limit. One 360 kW connector is shared at the selected depot. All used buses start full and return full by a synthetic 30:00 service-block boundary. Charging has no taper and no switching time. This finite block does not establish a repeating daily operation. A 37-bus one-service-per-bus construction confirms feasibility of both cases, but has no optimization role. Monetary travel cost is zero: the objective charges the used-bus fee and grid energy, while movement energy and time remain in the physical constraints.

We first use synthetic used-bus cost $f = 100$ and flat grid price $a = 0.2$ per kWh. Because this objective is linear in load, minimizing it over the hull of complete plans has the same value as minimizing it over physical plans; this flat-price experiment cannot exhibit a physical-hull gap. The original fixed-budget native runs returned physically replayed two-bus schedules in both depot cases without proving objective optimality. A separate, already archived round-0 solve within the later nonlinear planner used the same complete depot-15 physical case and a tangent objective that reduces exactly to the flat price. That call returned a *native numerical* optimum of approximately 408.53, with matching incumbent and lower bound under solver tolerances. This is a retrospective interpretation of one call, not a new flat-price experiment or a reclassification of the nonlinear

campaign. The numerical certificate does not identify the exact rational-model optimum. The following ideal stored-input certificates answer that distinct exact-arithmetic question.

For both depot graphs, a one-bus path is impossible. Chronologically forced adjacent services leave no admissible depot visit, while mandatory service energy alone exceeds one bus’s 400 kWh inventory. An exact cardinality-gated path-cover relaxation of each full 37-service graph gives flat-objective lower bounds displayed outward as 404.92 at depot 15 and 414.39 at depot 16. The bounds retain every declared connection mode but relax some physical constraints. Separately, an exact rational depot-15 witness covers all 37 services with two buses, respects the shared connector, and restores both batteries to 400 kWh by 30:00. Hence depot 15’s ideal model has a minimum fleet of two and the *flat-cost optimum* lies in

$$D_{15}^{\text{flat}} \in [404.92, 408.54] \quad (\text{outward-rounded exact rational bounds}). \quad (13)$$

The bounds do not identify the optimum. Depot 16 has the exact at-least-two obstruction and lower bound, but its returned two-bus upper witness remains numerical, yielding a mixed-evidence interval $[414.39, 433.75]$ after outward rounding. In particular, a native solver’s rounded matrix and its feasibility tolerances are distinct from the ideal stored-input rational model.

For a nonlinear illustration on *the same finite depot-15 ideal model*, let e_t be hourly grid energy for 30 market intervals and set

$$F(e) = \sum_{t=1}^{30} \left(0.2e_t + \frac{e_t^2}{1800} \right). \quad (14)$$

The coefficients and bus cost remain synthetic. The exact two-bus witness gives a physical upper bound displayed as 512.77. To bound the complete hull from below, let L_{flow} denote the exact depot-15 two-path flat-cost relaxation value. Because monetary travel cost is zero, it implies at least $E_{\text{min}} = (L_{\text{flow}} - 2f)/a \approx 1024.62$ kWh for any two-bus plan. One bus is impossible, and a uniform-price comparison using the service-energy floor excludes three-or-more-bus plans from improving the dual lower certificate. The 30-period Fenchel inequality then gives the exact rational lower bound displayed as 424.36. Therefore

$$424.36 \leq CH_{15} \leq D_{15} \leq 512.77, \quad 0 \leq D_{15} - CH_{15} \leq 88.41. \quad (15)$$

All displayed endpoints are rounded outward from the exact bounds. They do not establish CH_{15} or D_{15} individually, nor a strictly positive gap. The public case thus demonstrates physically feasible complete-fleet witnesses and useful bounds under declared assumptions, while leaving public price support open.

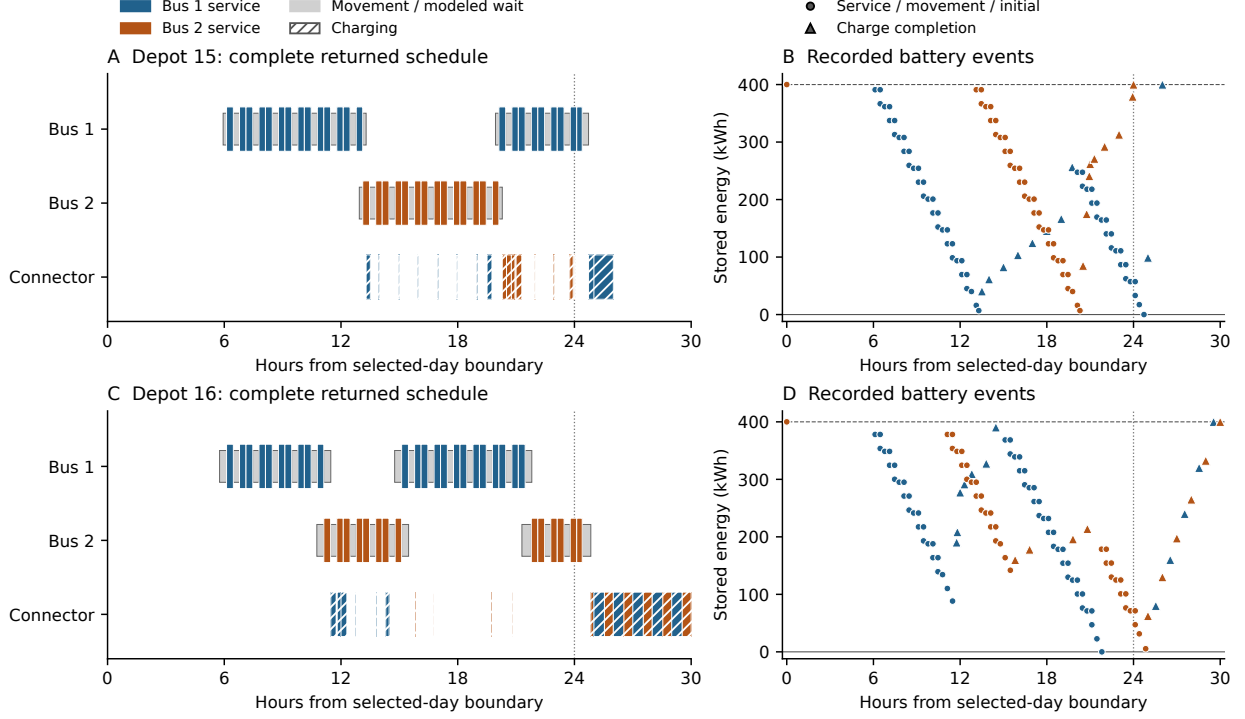


Figure 3: Two-bus schedules returned by the original flat-price pilot for the two depot variants of the same Hildenbrand timetable. Battery and charging traces describe numerical feasible witnesses under the declared scenario assumptions. They are not measured operation, two independent networks, or proofs of cost optimality. The separately constructed exact depot-15 witness supplies the ideal-model bounds in the text.

6 Computational evidence

The computations ask whether complete-fleet convexification can be bounded at a changed supply market, and which parts of repeated optimization remain costly. They do not estimate a population-level speedup. We use two deliberately small synthetic cases and two depot variants of a single 37-service Hildenbrand timetable. Depot 15 and depot 16 use different depots in that base network; they are not independent public replications. The reported results come from one run of each campaign. The campaigns reuse cases, code paths, and in some instances source-plan evidence, so their cells cannot be pooled as independent trials.

For a case I , let $\mathcal{F}(I)$ be its set of *complete* physically feasible fleet schedules. A column $x \in \mathcal{F}(I)$ includes the service assignment, vehicle paths, charging decisions, and full-horizon load $e(x)$, with operating cost $c(x)$. For the separable quadratic supply model

$$G_s(e) = \sum_{t=1}^T \left(a_{s,t} e_t + \frac{1}{2} b_t e_t^2 \right), \quad s \in \{0, 1\}, \quad (16)$$

the pure-fleet and complete-fleet-hull objectives are

$$\begin{aligned} D_s &= \min_{x \in \mathcal{F}(I)} \{c(x) + G_s(e(x))\}, \\ CH_s &= \min_{(e,c) \in \text{conv}\{(e(x), c(x)) : x \in \mathcal{F}(I)\}} \{c + G_s(e)\}. \end{aligned} \quad (17)$$

Table 1: Development campaigns and their declared scope. Outcomes count cells for the two iterative-hull campaigns and fixed-price calls for the pricing-start campaign. The same four physical cases recur across campaigns.

Campaign	Design	Observed outcomes
Feasible-pool pilot	Three hull methods $\times$ four cases $\times$ two markets; reserve and source-column reuse	24 cells: 7 certified and 17 budget-exhausted. Both public target depot variants remained open.
Solver-baseline comparison	Four ordered hull methods $\times$ four cases $\times$ two markets; numerical restricted master and physical-bound cache	32 cells: 11 certified, 18 budget-exhausted, 3 failed. No public target cell certified.
Pricing-start pilot	Cold versus submitted movement start at two fixed price queries on four cases	16 calls returned. Four synthetic pairs agreed within tolerance; all eight public calls reached the native 160-s cap.

State 0 is the source market and state 1 is the changed target market on the *same* physical case. In the public cases $T = 30$, $b_t = 1/900$, and $a_{0,t} = 0.20$ throughout; the target has $a_{1,t} = 0.18$ in the first 15 periods and 0.22 in the last 15. The synthetic cases have four periods and similarly paired source and target coefficients. This controlled price change makes same-case reuse meaningful without treating a source-market solution as a target-market optimum.

We report an interval $[L, U]$ for CH_1 only when the run produced the required evidence. Its upper endpoint is the objective of a feasible *convex mixture* of complete fleet columns. That mixture is a valid hull point, but generally cannot be dispatched as one fleet schedule. A global lower endpoint is assembled from physical fleet-pricing evidence in the target market, including checked inherited physical-pricing bounds where explicitly indicated. Native mixed-integer solver bounds and replayed numerical solutions retain the solver’s tolerance qualifications; they are computational enclosures, not exact proofs for the ideal model. A zero or small gap in the finite *restricted* column pool does not close the global interval: ungenerated complete fleets still matter. Nor does an interval for CH_1 alone establish the pure-fleet value D_1 , a nonconvexity gap $D_1 - CH_1$, or support by any proposed tariff.

6.1 Complete-fleet hull updates

The first pilot compared a legacy cold run, a cold run that reserves time for a subsequent restricted master, and that reserve policy with feasible source-column reuse. The reserve made a second master possible in all four public reserve-cold cells, while the corresponding legacy cells ended after one master. In the changed market, the reuse method admitted two feasible state-0 columns to its first master, obtained one fresh target-market priced column, and processed it in a second master. The source columns help form target-market feasible mixtures, but they do not replace target-market pricing for a global lower bound.

On the synthetic cyclic case, all six market-by-method cells certified. On the multivisit case, the source market exhausted its budget in all three methods. In the target market, reuse closed the global enclosure while the two cold methods exhausted their budgets: its displayed interval is $[84.72, 84.73]$ after outward rounding, versus $[81.17, 86.05]$ for reserve cold. This is a diagnostic of the method under the declared small-case budget, not a claim about larger networks.

Table 2: Public target-market hull evidence in the ordered solver-baseline comparison. L is rounded down and U up to two decimals; unrounded stored values determine certification. “Exhausted” means budget exhausted. Paid time includes both market states and, where applicable, parent source admission. A failed target has no final interval even though its time was paid.

Depot	Method	L	U	Paid (s)	Outcome
15	Reserve cold	—	—	347.20	failed
15	Reserve + pool	315.76	469.89	354.61	exhausted
15	QP + pool	—	—	349.88	failed
15	QP + pool + cache	405.83	471.52	352.27	exhausted
16	Reserve cold	377.65	531.90	351.08	exhausted
16	Reserve + pool	350.87	490.67	353.48	exhausted
16	QP + pool	—	—	349.53	failed
16	QP + pool + cache	420.45	491.46	352.22	exhausted

For the public target cases, reserve cold lowered the feasible upper relative to legacy cold, from 547.47 to 526.31 at depot 15 and from 549.70 to 531.90 at depot 16. Reuse lowered those upper endpoints further to 469.89 and 490.67. Yet its fresh global lower endpoints were 315.76 and 329.22, below reserve cold’s 339.35 and 377.65. Thus the upper improvements of 56.43 and 41.24 do not imply tighter *global* enclosures; in depot 16 the interval width actually grows. The public reuse cells stopped at projected rational-polishing bit growth, whereas reserve-cold cells stopped at the explicit pricing-reserve guard.

The ordered solver-baseline comparison then retained feasible columns and introduced two further mechanisms. A numerical quadratic-program (QP) *restricted-master proposal* selects mixture weights over the currently known complete-fleet columns; a checked rational replay admits its objective and residual. The fourth method also re-evaluates an inherited *physical-pricing lower bound* under the changed market’s Fenchel conjugate. This cache concerns already-checked physical pricing evidence, not a cache of an unverified schedule or a substitute for all fresh target-market pricing. Table 2 gives all public target outcomes in this comparison, including the paid source-state work and source-admission cost for retained methods.

The completed public QP-and-cache cells ended with restricted-pool residuals below the configured 10^{-6} pool tolerance, without exhausting the rational-polishing bit budget. Their *global* intervals were still open by 65.68 and 71.00 objective units, and both stopped on native remaining-time budget. In each, one fresh physical target pricing result was recorded. The inherited, target-revalued physical lower exceeded the best fresh lower within the same run by 98.82 at depot 15 and 69.83 at depot 16; the selected global lowers were 405.83 and 420.45. This within-run decomposition identifies a useful source of bound strength. It cannot isolate a causal cache effect across methods: the QP-without-cache target cells failed, and the cached method also incurs source-state computation and admission work.

An implementation error when a pricing call returned no incumbent caused three target failures: depot 15 reserve cold and both depot variants with the numerical restricted master and pool but no bound cache. No new completed pricing result or global bound was produced for the failing call. Their spent time remains in Table 2, and their missing intervals cannot be imputed from another method. On the two synthetic cases, all methods certified cyclic in both markets, while the three retained-pool methods certified multivisit in the changed market and reserve cold did not.

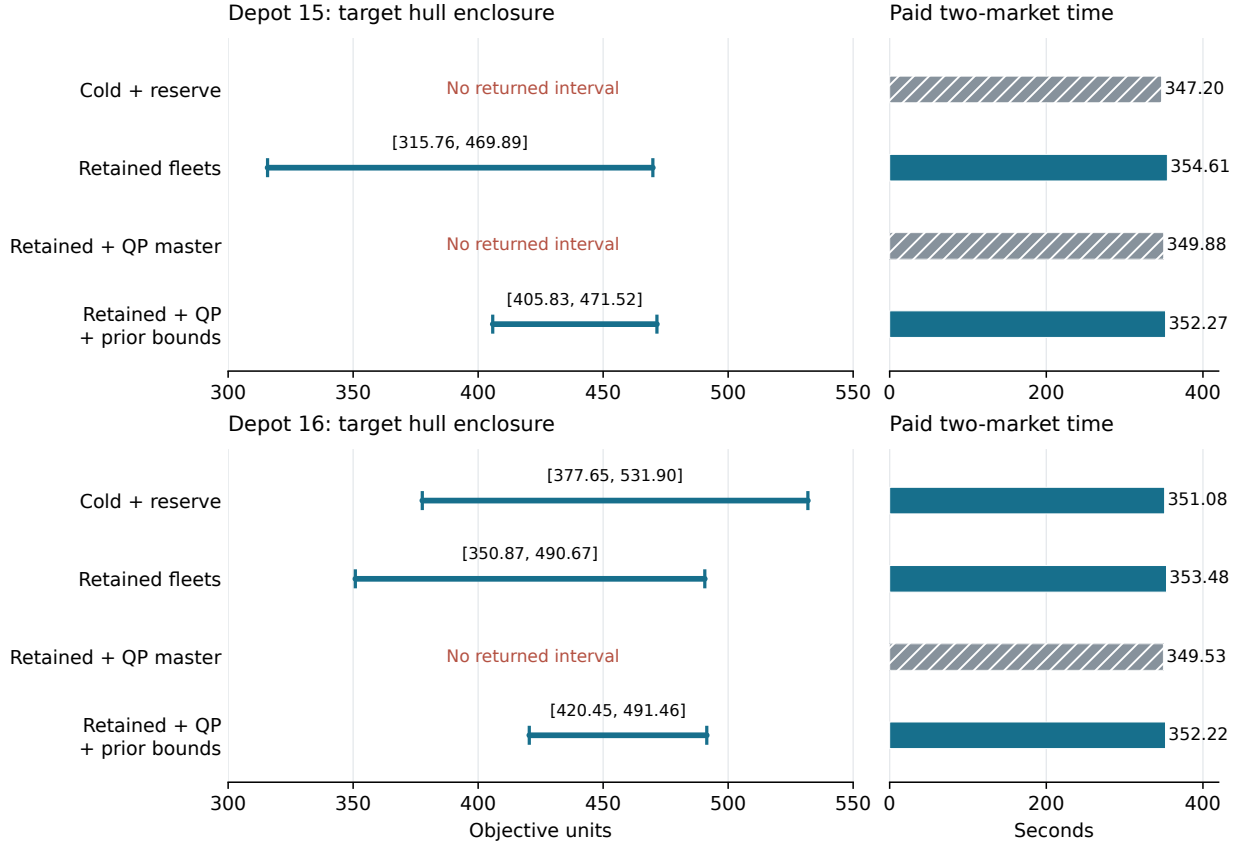


Figure 4: All eight public target cells in the ordered solver-baseline comparison: global hull intervals and paid two-market cost. Lower endpoints use physical-pricing evidence and upper endpoints feasible mixtures. Every completed interval remains open. Hatching marks a failed target with no final bound interval; its spent time remains in the paid-cost panel. The depot variants share one timetable.

The iterative timings explain why a faster restricted master alone would have limited impact on these public cells. Recorded successful public physical-pricing calls were about 160 s, while QP proposal took 0.77–1.07 s and rational replay 0.51–0.53 s in the completed cached targets. The feasible-pool pilot’s public changed-market pricing consumed roughly 160–172 s per cell; its master native solves were below 0.01 s, though rational polishing in reuse took about six seconds. Component scopes differ and may overlap, so these quantities are not an additive time budget. In the first pilot, paid source-plus-target times were about 369 s for legacy cold, 350 s for reserve cold, and 351–352 s for reuse on the public variants. Those methods ended at different bounds and stopping points. Neither campaign specified an equal-quality time target, and none of these differences is an equal-quality speedup estimate.

6.2 Fixed-price movement starts

The separate pricing-start pilot probes the physical pricing oracle at fixed prices rather than running the iterative hull. Its *linear* query uses the target market’s coefficients $p_t = a_{1,t}$. Its *marginal* query uses $p_t = a_{1,t} + b_t e_t(\bar{x})$, where $\bar{x}$ is the cheapest linear-bill schedule among the common feasible source columns. This anchor is a selected feasible column, not a certified target planner optimum. The two price vectors are distinct in all four cases. A common feasible source schedule supplied a

Table 3: Public fixed-price pricing at the same native 160-s cap. Each pair has a common feasible source upper. The displayed lower is the admitted native lower (rounded down); the displayed upper is the physically replayed native incumbent (rounded up). “Paid” is the complete child time for cold/start, not just native optimization.

Depot	Price	Cold L	Cold U	Start L	Start U	Paid cold/start (s)
15	linear	345.86	413.67	340.18	413.67	163.01 / 163.31
15	marginal	383.00	419.06	379.41	417.95	163.15 / 163.25
16	linear	377.65	430.86	377.65	430.86	163.11 / 163.20
16	marginal	416.72	438.17	428.51	438.17	163.11 / 163.17

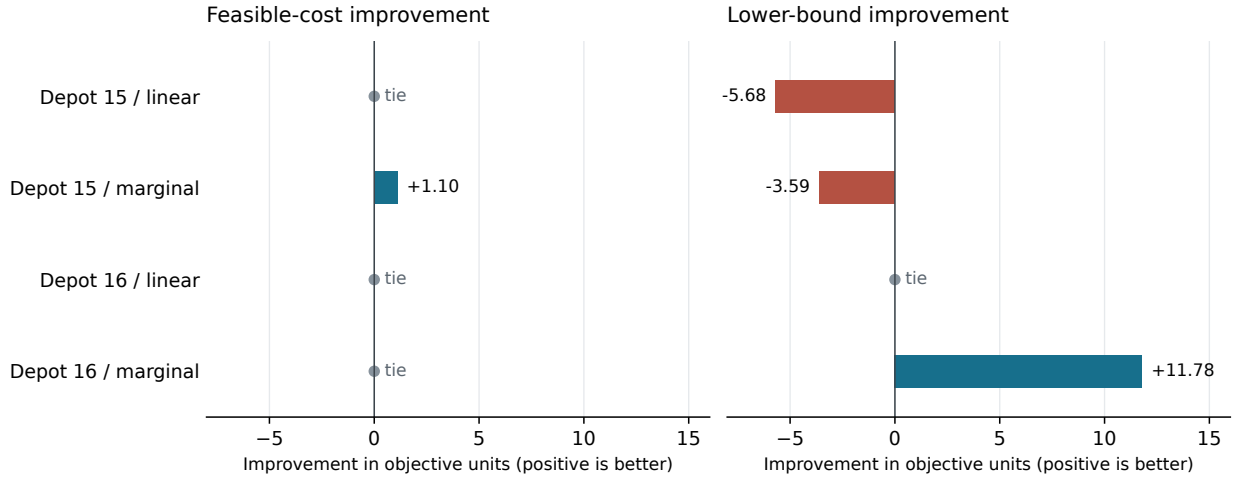


Figure 5: Within-pair public effects of a submitted movement start on replayed incumbent and admitted lower bound at fixed native caps. A smaller incumbent and a larger lower bound are favorable; the effects differ by query and depot.

baseline upper to both arms; the start arm additionally submitted its checked discrete movement selections to the native solver, leaving continuous charging as a solver decision. Submission was observed, but native acceptance or use of the hint was not.

The four synthetic pairs returned the same certified numerical intervals under cold and submitted-start calls, within the comparison tolerance. Table 3 reports the four public pairs. Every public native call ran to the 160-s cap and returned a physically replayed incumbent. On depot 15’s marginal query the start improved that incumbent by 1.10 but worsened the admitted lower by 3.59. On depot 16’s marginal query the incumbent tied while the admitted lower improved by 11.78. The depot 15 linear lower worsened by 5.68, and the depot 16 linear pair tied. These crossed upper/lower movements give no consistent bound-quality improvement.

The historical construction of the four shared source pools cost 358.21 s once. The preparation command, including source admission, cost a further 1.96 s. The 16 complete child receipts sum to 1,317.51 s; adding the once-paid source work and freeze command gives 1,677.67 s for this source-inclusive accounting. Nested native, setup, and child timings must not be added again. The source schedule was only modestly worse than the best newly returned public incumbent at this cap, with excess ranging from 0.32% to 2.74% of that returned incumbent’s fixed-price objective. This denominator is a feasible result, not the unknown fixed-price optimum. No time to first incumbent

Table 4: Evidence map for the computational claims. “Established” refers to the declared development cells and native-tolerance-qualified intervals, not to independent statistical replication.

Established in these cells	Still untested or unresolved
Complete-fleet feasible columns can be reused across the two markets, and the multivisit target enclosure closed under reuse.	Whether reuse improves target-market global gaps or paid time across independent public timetables at equal quality.
Checked inherited physical-pricing evidence strengthened the selected lower in two completed public cached runs.	A causal cache effect against a completed, equally budgeted QP-without-cache target and a public global certification.
Movement starts were submitted and public fixed-price calls returned mixed incumbent/lower effects at equal native caps.	Native hint acceptance, time to first incumbent, and any robust start-induced speedup or learned-proposal benefit.

was recorded, so the observed equal-cap quality cannot be converted into a start-induced time-to-solution claim.

6.3 What the experiments establish

The open public global intervals therefore remain the central computational limit for price support: a nearly solved restricted pool does not determine the full fleet hull or certify a target-market tariff. Nearest-neighbor retrieval and learned proposals remain untested here; evaluation across independent timetable groups is future work.

7 Implications and limits

The analytical and computational parts answer complementary questions. The construction proves that continuous charging and full replenishment do not remove the possibility of a marginal-price support failure. The experiments measure how difficult it is to obtain informative full-fleet bounds on declared instances. An unresolved public interval neither confirms the analytical mechanism on that timetable nor refutes it. Likewise, a strong restricted-master solution says little about the cost of discovering an ungenerated physical fleet unless global pricing has supplied a corresponding bound.

This distinction informs the route toward faster online decisions. Retained complete fleets already provide feasible candidates after a price-only change when the physical case remains unchanged. A nearest-neighbor proposal or a learned route predictor should first be compared against that baseline and against choosing the cheapest current bill from the same stored pool. Route fragments require assembly and physical repair before becoming a complete feasible fleet. Evaluation should report direct proposal quality, repair effort and optional global verification separately; a good dispatch need not come with a global optimality certificate. Training-data generation and source optimization are costs, even when amortized over later queries.

The present studies do not evaluate those retrieval or learned methods. They also do not establish transfer across timetables, demand regimes or charger designs. Future comparisons must keep all depot, tariff and timing variants of a base timetable within one data group and reserve independent timetables before training. Time-limited incumbents should retain their bound gaps instead of becoming exact labels. Repeated seeds, independent cases and time-to-common-quality targets would be needed before interpreting runtime differences as systematic acceleration.

The physical model abstracts from stochastic delays, battery degradation, temperature-dependent consumption, nonlinear charging taper and operational switching times. Public service and travel inputs are combined with declared energy and economic assumptions; the results are not estimates of savings for the source operator. Resource rights also delimit the economic interpretation. A whole-fleet price-taking deviation, an individually reserved-pair deviation and a strategic operator anticipating its own price impact are different optimization problems. Conclusions must follow the chosen participant and settlement model.

8 Conclusion

A smooth convex electricity supply cost need not support an indivisible electric-bus fleet at its own marginal prices. A fully replenished construction exhibits this failure and retains it under a specified robust variant. Replication can make the planning gap small without making the incentive of one whole-fleet operator small along every subsequence, while incentives for reserved pairs vanish. Development computations show that feasible-fleet reuse and inherited pricing bounds can help particular solves, but fresh global pricing remains costly and public gaps remain open. The current evidence supports further controlled computational work; it does not yet establish a learned method or general online acceleration.

Data and reproducibility

The EGG research repository contains the analytical constructions, declared public-data transformation, experiment protocols, complete outcome tables including unsuccessful attempts, and figure-generation sources. A companion source map links the manuscript’s claims to reviewed evidence and distinguishes exact ideal-model calculations from tolerance-qualified native results. All cases reported here are development cases. No private operational data or reserved evaluation outcomes are used. The figures in this manuscript are drawn from already reviewed results; its assembly required no new optimization. The supplementary research record retains execution and qualification details without treating those checks as empirical performance results.

A Proofs for the complete-fleet statements

Proof of Proposition 1. For physical x and any hull point (e, c) , linear minimization over the convex hull gives $c + p_x^\top e \geq V(p_x)$. Convexity gives $F(e) \geq F(e(x)) + p_x^\top (e - e(x))$. Adding and minimizing over the hull yields $CH \geq c(x) + F(e(x)) - r(x)$, proving the first inequality; $c(x) + F(e(x)) \geq D \geq CH$ proves the second. If $r(x) = 0$, all inequalities collapse to $CH = D = c(x) + F(e(x))$. Conversely, if $CH = D$, compactness supplies a physical minimizer x^* . It is also a hull minimizer. For every physical y , the directional derivative of $c + F(e)$ at $(e(x^*), c(x^*))$ toward $(e(y), c(y))$ is

nonnegative: $c(y) + p_{x^*}^\top e(y) \geq c(x^*) + p_{x^*}^\top e(x^*)$. Thus $r(x^*) = 0$. The same argument applies to every physical minimizer. $\square$

For the accounting identity, $F^*(p) \geq p^\top e(x) - F(e(x))$ by definition, so supplier LOC is non-negative. Substitution in (4) gives the claimed sum. A compact hull and a supply cost differentiable near it yield a supporting price p^* at a hull minimizer (e^*, c^*) ; linear first-order optimality gives $V(p^*) = c^* + (p^*)^\top e^*$, and Fenchel equality gives $F^*(p^*) = (p^*)^\top e^* - F(e^*)$. Hence $V(p^*) - F^*(p^*) = CH$. At a physical planner optimizer the LOC sum is $D - CH$. At the plan's own gradient price, Fenchel equality makes supplier LOC zero.

Proof of (6). Let (e^*, c^*) be a hull minimizer, $p^* = a + Qe^*$, $d = \|e(x) - e^*\|_Q$, and $H = H(x)$. First-order hull optimality makes (e^*, c^*) a minimizer of $c + (p^*)^\top e$, so

$$\text{LOC}_{\text{fleet}}(x; p^*) = H - \frac{1}{2}d^2 \geq 0.$$

In particular $d \leq \sqrt{2H}$. For each physical alternative y , the change in its linear advantage when the price moves from p^* to $p_x = p^* + Q(e(x) - e^*)$ is $(e(x) - e^*)^\top Q(e(x) - e(y))$, at most dR_Q by the seminorm Cauchy–Schwarz inequality. Maximizing over y gives $r(x) \leq H - d^2/2 + dR_Q \leq H + R_Q\sqrt{2H}$. The lower inequality is Proposition 1. $\square$

Proof of Proposition 2. Let λ be the one-bus mixture weight and x the mean early load. Complete-plan mixtures have $0 \leq \lambda \leq 1$, $10\lambda \leq x \leq 10$, and intrinsic cost $14 - 7\lambda$. These conditions are sufficient: for $\lambda < 1$ the two-bus component buys $(x - 10\lambda)/(1 - \lambda) \in [0, 10]$ early; at $\lambda = 1$, $x = 10$. For fixed x , cheapest intrinsic cost uses $\lambda = x/10$. Substituting $z = 30 - x$ into (7) gives hull objective

$$14 - \frac{7x}{10} + F(x, 30 - x) = 104 - 2.7x + 0.2x^2 = \frac{7591}{80} + \frac{(x - 27/4)^2}{5}.$$

Its minimizer is $x = 27/4$, $\lambda = 27/40$. Direct minimization of the one- and two-bus physical branches gives respectively 97 and 99. At $(10, 20)$ the gradient of F is $(6, 4)$; its private objective is $7 + 6(10) + 4(20) = 147$, while the two-bus $x = 0$ plan has $14 + 4(30) = 134$. At hull load $(27/4, 93/4)$ the gradient is $(107/20, 93/20) = (5.35, 4.65)$. Both supporting plans have private objective 153.5, and $F^*(p) = 58.6125$ on $\mathbb{R}_+^2$. The hull dual value is therefore 94.8875. The physical one-bus plan is a price minimizer, so fleet LOC is zero; supplier LOC is $97 - 94.8875$. $\square$

Proof of Proposition 3. For reserve $0 \leq r \leq 5$, efficiency η , one-hour early limit P , and one-hour single terminal connector K , write

$$T = 30/\eta, \quad \ell = \max\{0, T - K\}, \quad u = \min\{P, 15/\eta\}, \quad h = \max\{\ell, (10 + r)/\eta\}.$$

The two-bus physical early-energy interval is $[\ell, u]$, and the one-bus interval is $[h, u]$ when nonempty. The lower limit ℓ enforces terminal connector capacity, while $(10 + r)/\eta$ ensures the one bus can begin B above reserve. For every point in these intervals, charge only A's bus early; terminal demands total $T - x \leq K$ and can be served serially in the one-hour terminal window. Individual inventories and acceptance limits hold, and all used buses finish at capacity.

Write $G_T(x) = F(x, T - x)$, $k = 1/5$, and $m = T/2 - 5a/2$ for early linear coefficient $a = 4$. Then $G_T(x) = G_T(m) + k(x - m)^2$. Suppose $\ell < m < h < u$, $f > 0$, and $f < 2k(h - \ell)(h - m)$. The cheapest hull operating-cost boundary over $[\ell, h]$ is $2f - f(x - \ell)/(h - \ell)$, obtained by mixing the

two-bus endpoint ℓ and one-bus endpoint h ; over $[h, u]$ it is f . Put $d = h - \ell$. The strict inequality places the hull minimum $x^* = m + f/(2kd)$ inside (m, h) . Physical branch minimization gives

$$D = G_T(m) + \min\{f + k(h - m)^2, 2f\}, \quad CH = G_T(m) + 2f - \frac{f(m - \ell)}{d} - \frac{f^2}{4kd^2}.$$

Consequently

$$\Delta = \min \left\{ k(h - x^*)^2, \frac{f(m - \ell)}{d} + \frac{f^2}{4kd^2} \right\} > 0.$$

At $(r, \eta, P, K, f, a) = (1, 19/20, 12, 30, 7, 4)$, $(\ell, m, h, u, d) = (30/19, 110/19, 220/19, 12, 10)$. The inequalities are strict: $u - h = 8/19$ and $2kd(h - m) = 440/19 > 7$. Rational substitution yields (9). All strict inequalities persist under small changes in (r, η, P) with $K = 30$ fixed. $\square$

Proof of Proposition 4. For m paired buses, total early energy E ranges over $[10m, 10n]$, with intrinsic cost $14n - 7m$ and terminal energy $30n - E$. Convexifying complete plans gives a triangular projected set; its cheapest operating cost at each E is $14n - 0.7E$. Substitution in (10) gives a quadratic minimized at $E = 27n/4$ with value $7591n/80$.

For fixed integer $m \geq n/2$, physical supply cost is minimized at $E = 10m$, and the total objective is $104n - 27m + 20m^2/n = 7591n/80 + 20(m - 27n/40)^2/n$. For $m \leq n/2$, the best early load is $E = 5n$ when feasible and the branch objective is $99n - 7m$, decreasing toward $n/2$. For odd n , moving from $m = \lfloor n/2 \rfloor$ to $m = \lceil n/2 \rceil$ changes the minimized objective by $-7 + 5/n < 0$. For even n , the two formulas agree at $m = n/2$, and an integer nearest to $27n/40$ lies in the upper regime. Thus the nearest integer to $27n/40$ is optimal, with both integers at a half-integer tie and $|\delta| \leq 1/2$.

At a physical optimum, the price is $p = (4 + 2m/n, 6 - 2m/n)$. The one-bus private objective minus the best two-bus private objective is $40\delta/n$; the latter buys zero early energy when $m/n \geq 1/2$. Hence whole-operator regret is

$$r_n = \begin{cases} m 40\delta/n, & \delta \geq 0, \\ (n - m)(-40\delta/n), & \delta < 0. \end{cases}$$

For $n = 40k + 1$, $m = 27k + 1$ and $\delta = 13/40$, giving (12). The same difference is a reserved pair's individual regret when its current choice is worse; $|40\delta/n| \leq 20/n$. Summing the disadvantaged pairs reproduces the whole-operator expression because each pair has its own reserved connector rights. $\square$

References

- George B. Dantzig and D. R. Fulkerson. Minimizing the number of tankers to meet a fixed schedule. *Naval Research Logistics Quarterly*, 1(3):217–222, 1954. doi: 10.1002/nav.3800010309. URL <https://onlinelibrary.wiley.com/doi/10.1002/nav.3800010309>.
- Marelot H. de Vos, Rolf N. van Lieshout, and Twan Dollevoet. Electric vehicle scheduling in public transit with capacitated charging stations. *Transportation Science*, 58(2):279–294, 2024. doi: 10.1287/trsc.2022.0253. URL <https://doi.org/10.1287/trsc.2022.0253>. Published online 12 December 2023.
- Lingwen Gan, Ufuk Topcu, and Steven H. Low. Optimal decentralized protocol for electric vehicle charging. *IEEE Transactions on Power Systems*, 28(2):940–951, 2013. doi: 10.1109/TPWRS.2012.2210288. URL <https://doi.org/10.1109/TPWRS.2012.2210288>.

- Paul R. Gribik, William W. Hogan, and Susan L. Pope. Market-clearing electricity prices and energy uplift, December 2007. URL https://www.lmpmarketdesign.com/papers/Gribik_Hogan_Pope_Price_Uplift_123107.pdf. Working paper, 31 December 2007.
- Bowen Hua and Ross Baldick. A convex primal formulation for convex hull pricing. *IEEE Transactions on Power Systems*, 32(5):3814–3823, 2017. doi: 10.1109/TPWRS.2016.2637718. URL <https://doi.org/10.1109/TPWRS.2016.2637718>.
- Richard P. O’Neill, Paul M. Sotkiewicz, Benjamin F. Hobbs, Michael H. Rothkopf, and William R. Stewart. Efficient market-clearing prices in markets with nonconvexities. *European Journal of Operational Research*, 164(1):269–285, 2005. doi: 10.1016/j.ejor.2003.12.011. URL <https://doi.org/10.1016/j.ejor.2003.12.011>.
- Axel Parmentier, Rafael Martinelli, and Thibaut Vidal. Electric vehicle fleets: Scalable route and recharge scheduling through column generation. *Transportation Science*, 57(3):631–646, 2023. doi: 10.1287/trsc.2023.1199. URL <https://doi.org/10.1287/trsc.2023.1199>.
- Dane A. Schiro, Tongxin Zheng, Feng Zhao, and Eugene Litvinov. Convex hull pricing in electricity markets: Formulation, analysis, and implementation challenges. *IEEE Transactions on Power Systems*, 31(5):4068–4075, 2016. doi: 10.1109/TPWRS.2015.2486380. URL <https://doi.org/10.1109/TPWRS.2015.2486380>.
- Hubert Maximilian Sistig, Philipp Sinhuber, Matthias Rogge, and Dirk Uwe Sauer. Dataset for evaluating costs and operations of public bus fleet electrification, 2025a. URL <https://doi.org/10.6084/m9.figshare.26088190.v1>. Version 1; CC BY 4.0.
- Hubert Maximilian Sistig, Philipp Sinhuber, Matthias Rogge, and Dirk Uwe Sauer. Evaluating costs and operations of public bus fleet electrification. *npj Sustainable Mobility and Transport*, 2:15, 2025b. doi: 10.1038/s44333-025-00030-y. URL <https://doi.org/10.1038/s44333-025-00030-y>.
- Ross M. Starr. Quasi-equilibria in markets with non-convex preferences. *Econometrica*, 37(1):25–38, 1969. doi: 10.2307/1909201. URL <https://doi.org/10.2307/1909201>.
- Weitiao Wu, Yue Lin, Ronghui Liu, and Wenzhou Jin. The multi-depot electric vehicle scheduling problem with power grid characteristics. *Transportation Research Part B: Methodological*, 155: 322–347, 2022. doi: 10.1016/j.trb.2021.11.007. URL <https://doi.org/10.1016/j.trb.2021.11.007>.